BOLT AND FIXING SCHEDULE

KEY	DESCRIPTION	QTY	UNIT	WHERE	FLAG
M16x50 8.8 bolt set	M16x50 grade 8.8 bolt + nut + 2 washers, hot-dip galvanised	10	sets	A, B, C	
M12x90 8.8 bolt set (timber)	M12x90 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); head + washer counterbored Ø60 x 13 into the packer face	8	sets	SB1	TLS
M12x130 8.8 bolt set (timber)	M12x130 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); head + washer counterbored Ø60 x 13 into the binder face	9	sets	SB2	TLS
M12x50 coach screw + washer	M12x50 coach screw + 36 washer, zinc; pre-drill Ø12 shank / Ø8 thread	10	no.	SB2	TLS
M12x100 coach screw + washer	M12x100 coach screw + 36 washer, zinc; pre-drill Ø12 shank / Ø8 thread	2	no.	SB2	TLS
M12 8.8 threaded rod 165 + 2 nuts (timber jamb)	M12 grade 8.8 threaded rod cut to 165, 2 nuts + 2 no. 40x40x4 timber washers, HDG (BS EN ISO 10684); both ends in Ø60 x 18 counterbores in the jamb plates (nuts below the plate faces); snug tight	3	sets	SC1	TLS
M12 8.8 threaded rod 173 + 2 nuts (timber jamb)	M12 grade 8.8 threaded rod cut to 173, 2 nuts + 2 no. 40x40x4 timber washers, HDG (BS EN ISO 10684); both ends in Ø60 x 18 counterbores in the jamb plates (nuts below the plate faces); snug tight	6	sets	SC2, SC3	TLS
M12x160 8.8 bolt set (timber)	M12x160 grade 8.8 bolt + nut + 40x40x4 timber washer + steel washer, HDG (BS EN ISO 10684); nut in a Ø60 x 18 counterbore in the jamb plate; head on the door-side face (tell the door supplier); snug tight	6	sets	PC1, PC2	TLS
M12x55 stud welded to RHS + nut + washer	M12x55 grade 4.8 drawn-arc stud (BS EN ISO 13918), welded to the RHS top before galvanising, threads chased; HDG nut (BS EN ISO 10684) + 40x40x4 timber washer in a Ø60 x 16 counterbore in the head binder (nut below the binder top; studs midway between the GT1 trusses)	9	no.	SB4	TLS
M16x200 R-KEM II anchor set	M16x200 grade 8.8 threaded rod, HDG, + nut + washer, Rawlplug R-KEM II resin, Ø18 x 125 hole	20	sets	PC1, PC2, SC1, SC2, SC3	

FLAG TLS = TLS detail - approved by the engineer.

TIMBER KIT BOLTED TO THE STEEL (TLS SUPPLY)

ITEM	SECTION	GRADE	LENGTH	ON	PURPOSE	FLAG
SB1 packer 70x147 C24	70x147	C24	2559	SB1	west face packer (ripped 70 from 75), 21.8 proud of the flange tips, for Acorn's MT1 truss hangers; stops 85 clear of the Detail C bolts each end; Ø14 holes at 215 + 300n from its S end, 73.5 below its top (mid-height), counterbored Ø60 x 13 on the west face	TLS
SB2 packer 66x147 C24	66x147	C24	2866	SB2	east face packer (ripped 66 from 75), 2.25 proud of the flange tips; Ø14 holes at 215.7 + 300n from its S end, 73.5 below its top (mid-height)	TLS
SB2 binder 45x254 C24	45x254	C24	2709	SB2	joist binder: the 5 family-room joists hang off its face (Acorn hangers); Ø14 holes at 53.7 + 300n from its S end, 151.5 below its top, counterbored Ø60 x 13 on the east face	TLS
SB2 top plate 50x145 C24	50x145	C24	2905	SB2	sawn 50 ripped to 145: deck edge + FF-EX1 bearing; starts at the end of the top cope (y 9946), 30 clear of the south stone leaf	TLS
SC1 jamb plate 38x140 C16	38x140	C16	2462	SC1	W face jamb plate: the panel jamb stud nails to it	TLS
SC1 jamb plate 30x140 C24 ripped	30x140	C24	2462	SC1	E face jamb plate: the panel jamb stud nails to it	TLS
SC2 jamb plate 38x89 C16	38x89	C16	2462	SC2	W face jamb plate: the panel jamb stud nails to it	TLS
SC2 jamb plate 38x89 C16	38x89	C16	2462	SC2	E face jamb plate: the panel jamb stud nails to it	TLS
SC3 jamb plate 38x140 C16	38x140	C16	2462	SC3	S face jamb plate: the panel jamb stud nails to it	TLS
SC3 jamb plate 38x140 C16	38x140	C16	2462	SC3	N face jamb plate: the panel jamb stud nails to it	TLS
PC1 jamb plate 38x140 C16	38x140	C16	2400	PC1	jamb plate on the face against the return panel (to the underside of the post cap)	TLS
PC2 jamb plate 38x140 C16	38x140	C16	2400	PC2	jamb plate on the face against the return panel (to the underside of the post cap)	TLS
SB4 packer 14mm marine ply 140	14x140	marine ply	5540	SB4	packer under the head binder, over the RHS and the post caps (3 lengths from 2440 sheets, joints over the RHS)	TLS
SB4 head binder 38x140 C16 (a)	38x140	C16	2813	SB4	head binder over the steel (GT1 bearing)	TLS
SB4 head binder 38x140 C16 (b)	38x140	C16	2813	SB4	head binder over the steel (GT1 bearing)	TLS

CONSUMABLES (NOT MODELLED)

KEY	DESCRIPTION	QTY	UNIT	WHERE	FLAG
Shim set 25mm	HDG steel shims 50x50: per post 4 stacks of 10+5+5+3+1+1 (= 25, 1 mm to adjust), one at each mid-side under the SHS wall	5	post sets	SC1-3, PC1-2 bases (mid-sides under the SHS walls)	
R-KEM II 300ml	Rawlplug R-KEM II 300ml cartridge + mixer nozzles (20 holes Ø18x125 ~ 0.17 L + nozzle waste per base)	2	no.	post bases	
Grout 25kg	Non-shrink cementitious grout 25 kg (5 bases x 25 gap ~ 12 L)	2	bags	post bases	
Vent plugs Ø20	HDG drive-in plugs / caps for Ø20 galvanising vents	20	no.	posts + GP1	TLS
Vent plugs Ø12	HDG drive-in plugs for Ø12 boot lap vents	11	no.	BT1	TLS
DPC 150	DPC 150 wide on the SB1 / SB3 top flanges under the stone (2729 + 2126 + laps)	6	m	SB1, SB3	

STEEL WEIGHT SUMMARY

MARK	SECTION	GRADE	CUT LENGTH	QTY	SECTION KG	PLATES KG	TOTAL KG	COATING
SB1	UB203x102x23	S355JR	2729	1	63.0	0.8	63.9	HDG (EN ISO 1461)
SB2	UB203x133x25	S355JR	2955	1	74.2	0.0	74.2	Shop primer (C2)
SB3	UB203x102x23	S355JR	2126.3	1	49.1	0.8	49.9	HDG (EN ISO 1461)
SC1	SHS100x100x6.3	S355J2H	2768.2	1	50.4	10.1	60.4	HDG (EN ISO 1461)
SC2	SHS100x100x6.3	S355J2H	2768.2	1	50.4	8.9	59.3	HDG (EN ISO 1461)
SC3	SHS100x100x6.3	S355J2H	2768.2	1	50.4	10.0	60.4	HDG (EN ISO 1461)
SB4	RHS300x100x12.5	S355J2H	5300	1	383.2	122.2	505.4	HDG (EN ISO 1461)
PC1	SHS120x120x10	S355J2H	2510	1	84.6	10.6	95.2	HDG (EN ISO 1461)
PC2	SHS120x120x10	S355J2H	2510	1	84.6	10.6	95.2	HDG (EN ISO 1461)
TOTAL					**889.9**	**174.0**	**1063.9**	

NOTES

1. **APPROVED FOR CONSTRUCTION - the engineer approved the steelwork and the TLS details on 13.10.26; fabricate to these drawings.**
2. Bolt sets counted from the model (one token per set). Timber kit: TLS supply, TLS detail - approved by the engineer.
3. Weights: sections at nominal kg/m, plates at 7850 kg/m3 less holes.

REVISIONS

REV	DATE	DESCRIPTION
C03	11.10.26	Steel fabrication drawings - first issue
C04	13.10.26	Approved by the engineer 13.10.26 - issued for construction

TLS TIMBER FRAME DESIGN SUITE

TLS TIMBER FRAME

Design & Manufacture

Scan: 3D model + drawings

CLIENT	PROJECT
MP Adams	MP Adams Plot 4, Eckington

DRAWING TITLE

BOLT FIXING AND TIMBER SCHEDULE

Fixings, timber kit, steel weights

SCALE	SIZE	DATE	DRAWN
n/a	A3	13.10.26	TS

STATUS	DRAWING No.	REV
CONSTRUCTION	P4-ST-SCH	C04

Do not scale. All dimensions in mm. Check all dimensions on site.